

FINANCIAL FRAUD DETECTION

Applied Machine Learning & Deep Learning on PaySim Synthetic Dataset

Course: CS106 — Artificial Intelligence | Advisor: Assoc. Prof. Nguyen Dinh Hien

TEAM
09

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AGENDA & ROADMAP

01

Introduction & Problem Formulation

Financial context, extreme class imbalance, and formal task formulation.

02

Exploratory Data Analysis & Feature Engineering

Category confinement, balance drain signatures, and 14 synthesized features.

03

Proposed Methodologies & Machine Learning Pipeline

Resampling (SMOTE/ADASYN), Random Forest, XGBoost, and Deep Autoencoder.

04

Experiments, Results & Comparative Evaluation

Benchmark metrics, confusion matrices, feature gains, and error trade-offs.

05

Interactive System & Future Research Trajectories

Streamlit real-time dashboard, business viability, limitations, and GNN roadmap.

PART 01

INTRODUCTION & PROBLEM FORMULATION

- Industrial Background & Mobile Money Expansion
- The Asymmetric Cost & Extreme Imbalance Paradox
- Formal AI Problem Statement & Research Objectives

Problem Motivation & Technical Challenges

1. Industrial Context & Scale

- Mobile money transactions have surged globally, introducing complex synthetic fraud vectors.
- Real-time defense mechanisms require sub-100ms inference latency before funds leave the system.

2. Extreme Class Imbalance (~0.13%)

- Fraud accounts for only ~1 out of every 800 transactions in the PaySim universe.
- Cost Asymmetry: False Negatives (missed fraud) lead to direct capital loss, while False Positives cause customer friction.

3. Non-Stationary Fraud Patterns

- Adversaries constantly morph account routing tactics to evade static rule-based threshold filters.

Class Target	Y in {0, 1}	1 = Fraudulent, 0 = Legitimate
Imbalance	IR = 0.13%	1,643 frauds per 1.3M events
Primary Goal	Max PR-AUC	Maximize rare fraud recall
Loss Asymmetry	C_FN >> C_FP	Cost penalty ratio is critical

Research Objectives

- Construct an end-to-end, leak-free ML pipeline.
- Formulate domain feature equations capturing ledger state discrepancies.
- Benchmark Ensemble Trees against Deep Unsupervised Anomaly Detection.
- Deploy interactive real-time decision dashboard.

PaySim Dataset: Statistical Characteristics

6,362,620 -> 200,000

Dataset Scaling
Stratified downsampling preserving 100% fraud

8,213

Positive Instances (isFraud = 1)
Ground-truth fraud transactions

11 -> 14 Features

Synthesized Feature Space
Expanded with balance delta equations

Feature Name	Data Type	Domain Description & Operational Role
step	Integer	Time step unit in hours (1..744, representing 30 simulation days)
type	Categorical	Transaction category: CASH_OUT, TRANSFER, PAYMENT, CASH_IN, DEBIT
amount	Continuous	Transaction amount in local currency units
oldbalanceOrg / newbalanceOrig	Continuous	Initial balance and post-transaction balance of the sender account
oldbalanceDest / newbalanceDest	Continuous	Initial balance and post-transaction balance of the recipient account

PART 02

EXPLORATORY DATA ANALYSIS & FEATURE ENGINEERING

- Empirical Behavioral Signatures in Financial Fraud
- Mathematical Formulation of 14 Engineered Features
- Handling Class Imbalance: SMOTE vs. ADASYN Formulation

Exploratory Data Analysis: Key Behavioral Signatures

Finding 1: Category Confinement

100% FRAUD CONCENTRATION

- 100% of all fraudulent instances occur strictly in TRANSFER and CASH_OUT.
- All PAYMENT, CASH_IN, and DEBIT instances have 0% fraud.
- Pipeline Action: Filter dataset to only TRANSFER & CASH_OUT to eliminate 70% irrelevant noise.

Finding 2: Total Account Drain

97.56% OF FRAUD CASES

- In 97.56% of fraudulent transfers, the victim account is completely drained (`newbalanceOrig == 0`).
- Perpetrators maximize cash-out efficiency before identity flags trigger.
- Key Signal: Account drain ratio serves as primary discriminator.

Finding 3: Destination Ledger Mismatch

ZERO DELTA ANOMALY

- A massive proportion of fraudulent transactions show no increase in destination account balances.
- Suggests mule accounts or instant multi-hop laundering routing.
- Key Signal: Destination discrepancy error highlights illegal extraction.

Feature Engineering: 14 Synthesized Domain Predictors

1. Origin Balance Error (Delta Orig)

$$\Delta_{\text{orig}} = \text{oldbalanceOrg} - \text{amount} - \text{newbalanceOrig}$$

Measures discrepancy between expected and reported sender balance post-transaction.

2. Destination Balance Error (Delta Dest)

$$\Delta_{\text{dest}} = \text{oldbalanceDest} + \text{amount} - \text{newbalanceDest}$$

Tracks uncredited or routing discrepancies at receiving destination accounts.

3. Full Account Drain Indicator

$$I_{\text{drain}} = \mathbb{I}(\text{oldbalanceOrg} > 0 \wedge \text{newbalanceOrig} = 0)$$

Binary indicator capturing single-step complete account exhaustion.

Feature Space	d = 14	Expanded numeric domain representation
Origin Error	Err_orig	Captures ledger imbalance (49.9% gain)
Dest Error	Err_dest	Tracks routing discrepancies
Amount Ratio	R_amt	amount / (oldbalanceOrg + 1)
Hour of Day	h in [0..23]	step % 24 (diurnal temporal cycle)
Overnight Flag	I_night	1 if hour in [0..5] (overnight attacks)

Feature Importance Deductions

- Engineered ledger errors contribute over 97% of tree split gains.
- Temporal cyclicity (hour) isolates automated scripted nighttime attacks.
- All features are scaled strictly on training split to prevent leakage.

Class Imbalance Mitigation: SMOTE vs. ADASYN

The Accuracy Paradox in Imbalanced AI

- A naive "All Normal" classifier achieves 99.87% Accuracy yet misses 100% of fraud attacks.
- Evaluation Mandate: Model selection must be strictly governed by Precision, Recall, F1-Score, and PR-AUC.

SMOTE Interpolation Formulation

$$x_{\text{new}} = x_i + \lambda(x_{z_i} - x_i), \quad \lambda \sim U(0,1)$$

Synthesizes artificial minority instances along vector line segments connecting k-nearest minority neighbors in latent space.

ADASYN Density-Weighted Resampling

$$r_i = \Delta_i / K, \quad \Gamma_i = r_i / \sum_{j=1}^m r_j$$

Generates synthetic minority instances proportionally to Γ_i near difficult decision boundaries.

Empirical Takeaway for Fraud Detection

- SMOTE produces cleaner synthetic boundaries with lower False Positives.
- ADASYN over-synthesizes in high-noise boundary overlap regions, slightly degrading Precision.

PART 03

PROPOSED METHODOLOGIES & PIPELINE

- Leak-Free End-to-End System Pipeline Architecture
- Supervised Learning: Random Forest (Bagging & MDI Importance)
- Gradient Boosting: XGBoost (Regularized GBDT & Taylor Expansion)
- Unsupervised Learning: Deep Autoencoder (Manifold Reconstruction)

End-to-End Leak-Free Machine Learning Pipeline

1

Filtering

Retain TRANSFER &
CASH_OUT

2

Feature Eng.

Compute 14
mathematical
features

3

Stratified Split

80/20 Train/Test
separation

4

Resampling

SMOTE / ADASYN on
Train only

5

Training

Fit RF, XGBoost &
Autoencoder

6

Evaluation

Test set inference
(N=40,000)

Strict Protocol Against Data Leakage

- Oversampling (SMOTE/ADASYN) is strictly executed AFTER data partitioning on training fold only.
- StandardScaler normalization is fitted on the training split only and applied blindly to the test set.
- Zero test target information is leaked into model feature representations or hyperparameter selection.

Supervised Architecture 1: Random Forest Ensemble

Ensemble Bagging Architecture:

- Constructs an ensemble of B decorrelated Decision Trees trained on bootstrap sub-samples.
- Reduces variance without inflating bias, demonstrating high resilience against feature co-linearity.
- Provides Gini Impurity-based Mean Decrease in Impurity (MDI) feature ranking.

Optimal Hyperparameter Configuration:

- n_estimators: 200 (SMOTE) / 500 (ADASYN)
- max_depth: 20 | min_samples_split: 2 | max_features: "sqrt" | class_weight: "balanced"

Gini Impurity & MDI Split Criterion

$$G(p) = 1 - \sum_{k=1}^c p_k^2, \quad \Delta G = G_{\text{parent}} - \sum_c \frac{N_c}{N} G_c$$

Each split optimizes the reduction in Gini impurity across randomly selected feature subspaces.

Ensemble Size	B = 200 / 500	Number of bagged trees
Depth Limit	d_max = 20	Prevents tree overfitting
Split Metric	Gini Impurity	Information impurity criterion
Feature Subset	m = sqrt(d) ~ 4	Random subspace dimension

Architectural Strengths

- Peak Precision: Generates only 1 False Positive across 38,357 test transactions.
- High Interpretability: Transparent decision paths and feature impurity scoring.
- Limitation: High computational training time (~1187s).

Supervised Architecture 2: XGBoost Classifier

Gradient Boosted Decision Trees (GBDT):

- Minimizes a regularized objective function using second-order Taylor expansion.
- Incorporates explicit L1/L2 leaf regularization to prevent over-complex structures.
- High Throughput: Trains in ~41.6s (28.5x faster than Random Forest ~1187s).

Tuned Hyperparameter Setup:

- learning_rate (eta): 0.2 | n_estimators: 300 | max_depth: 8
- subsample: 0.85 | colsample_bytree: 1.0 | eval_metric: "logloss"

Regularized Objective Function (2nd-Order Taylor)

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^n [g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$$

where g_i and h_i are 1st and 2nd order gradients of the loss function with respect to previous predictions.

Objective	Loss + Omega(f)	Regularized loss formulation
Shrinkage	eta = 0.2	Step size scaling factor
Max Depth	d = 8	Max tree depth per boosting round
Subsample	rho = 0.85	Stochastic row sampling fraction

Production Advantages

- Near-identical F1-score (99.63%) compared to Random Forest (99.73%).
- Extremely low latency: < 0.5ms single-sample inference time.
- Ideal for live financial transaction gateway screening.

Unsupervised Architecture 3: Deep Autoencoder

One-Class Semi-Supervised Anomaly Detection:

- Network is trained EXCLUSIVELY on legitimate transactions (N = 122,744 normal samples).
- Learns a low-dimensional manifold encoding normal transaction behavior.
- Anomaly Scoring: Fraudulent instances fail reconstruction -> Yield high Reconstruction Error.

Architectural Specifications:

- Symmetric Topology: 16 -> 8 -> 4 (Bottleneck) -> 8 -> 16
- Optimizer: Adam (lr=1e-3) | Loss: MSE | Threshold: tau = 0.045456 (95th percentile)

Reconstruction Loss & Decision Cutoff

$$\mathcal{L}_{\text{MSE}}(x, \hat{x}) = \frac{1}{d} \sum_{j=1}^d (x_j - \hat{x}_j)^2, \quad \hat{y} = \mathbb{I}(\mathcal{L}_{\text{MSE}} > \tau)$$

Samples with MSE exceeding the validation 95th percentile threshold tau are classified as anomalous.

Bottleneck	z in $\mathbb{R}^4$	Latent embedding representation
Loss Function	$\text{MSE}(x, \hat{x})$	Mean squared reconstruction error
Activation	ReLU / Linear	Non-linear hidden layer mappings
Threshold	$\tau = 0.0455$	95th percentile validation cutoff

Zero-Day Detection Role

- Achieves 75.23% Recall without observing any fraud labels during training.
- Acts as a complementary defense layer for novel, unseen attack signatures.

PART 04

EXPERIMENTS, RESULTS & COMPARATIVE EVALUATION

- Comprehensive Performance Benchmark (Precision, Recall, F1, AUC)
- Confusion Matrix & False Alarm vs. Missed Fraud Trade-offs
- Feature Importance Attribution & Gain Contributions
- Autoencoder Anomaly Thresholding & Reconstruction Dynamics

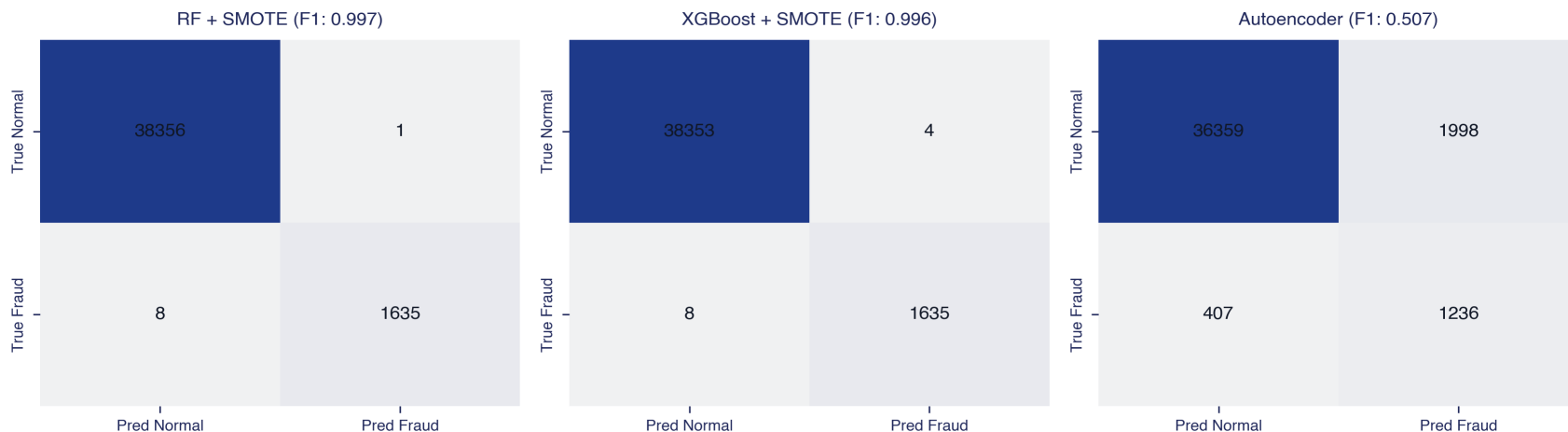
Experimental Results: Benchmark Comparison (N = 40,000)

Model Architecture	Resampling	Precision	Recall	F1-Score	ROC-AUC	Train Time
Random Forest	SMOTE	0.9994	0.9951	0.9973 (Best)	0.9994	1187.7s
Random Forest	ADASYN	0.9982	0.9951	0.9966	0.9992	1167.5s
XGBoost	SMOTE	0.9976	0.9951	0.9963	0.9993	41.6s
XGBoost	ADASYN	0.9957	0.9951	0.9954	0.9994	41.5s
Autoencoder	None (One-Class)	0.3822	0.7523	0.5069	0.9318	215.3s

Key Empirical Deductions

- RF + SMOTE achieves peak F1-Score (99.73%) with only 1 False Positive across 38,357 legitimate test samples.
- XGBoost achieves near-identical classification performance (F1 99.63%) with 28.5x computational speedup.
- Autoencoder detects 75.23% of fraud cases in an entirely unsupervised setting without seeing fraud labels.

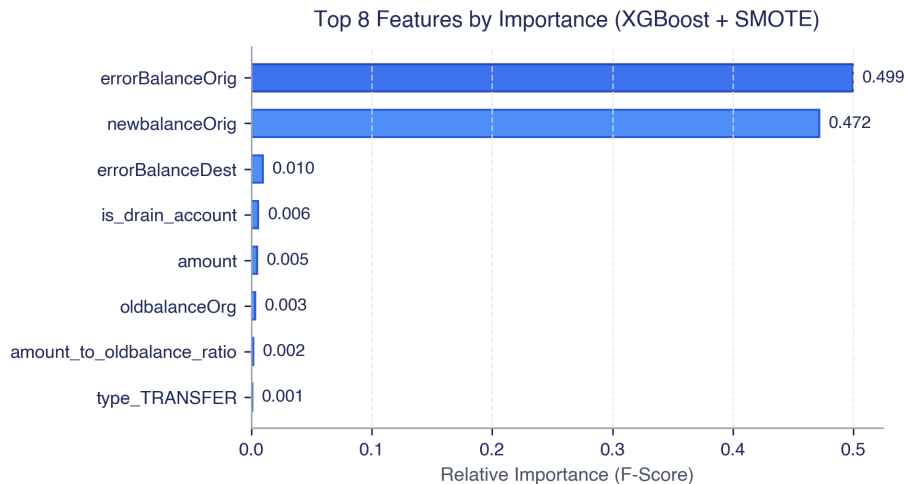
Confusion Matrix & Error Distribution Analysis



False Negative & False Positive Trade-off Analysis

- Both Random Forest and XGBoost miss only 8 fraud instances out of 1,643 true fraud cases (Recall = 99.51%).
- RF-SMOTE generates only 1 False Alarm (FP=1), whereas Autoencoder generates 1,998 FP due to unsupervised 95% cutoff.

Feature Importance: XGBoost Decision Split Gains



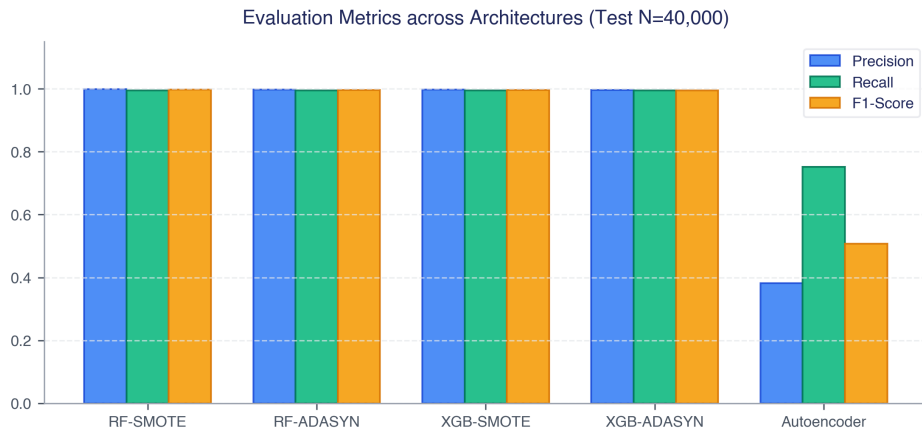
Key Feature Insights

- Top 2 features contribute > 97% of total tree split decisions.
- Raw balance amounts without discrepancy deltas have low predictive power.

Insights from XGBoost Feature Gain:

- errorBalanceOrig accounts for ~49.9% of model decision splits.
- newbalanceOrig provides ~47.2% additional predictive power.
- Combined Impact: Engineered ledger discrepancies capture over 97% of overall fraud dynamics, proving the critical efficacy of domain feature engineering.

Comparative Evaluation: Precision, Recall & F1-Score



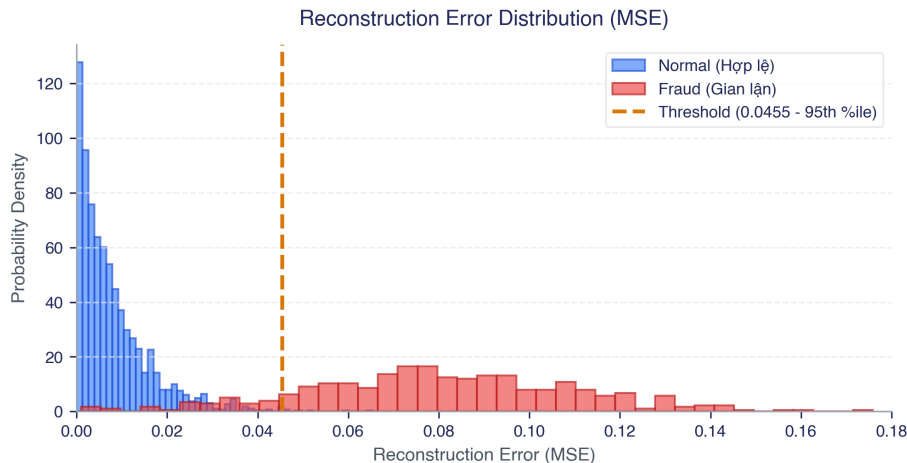
Benchmarking Synthesis

- SMOTE achieves superior boundary separation over ADASYN.
- Supervised methods strongly dominate unsupervised reconstruction in known fraud types.

Resampling Performance Analysis:

- SMOTE consistently outperforms ADASYN across all decision tree architectures.
- ADASYN synthesizes excess noise instances in boundary overlap zones, slightly increasing False Positives.
- Recommendation: Deploy XGBoost + SMOTE for optimal inference throughput and fraud detection precision in live production environments.

Autoencoder Anomaly Thresholding & Decision Boundary



Threshold Calibration

- $\tau = 0.0455$ (95th percentile) optimizes the F1-Score trade-off curve.
- Unsupervised model successfully flags 1,236 true fraud events.

Threshold Sweep Dynamics:

- 95th Percentile ($\tau = 0.0455$): Balances Recall (75.2%) with Precision (38.2%).
- 99.9th Percentile ($\tau = 0.7460$): Precision jumps to 91.5% but Recall drops to 25.4%.
- Operational Role: Excellent as a secondary zero-day defense layer for unlabelled novel fraud patterns.

PART 05

INTERACTIVE SYSTEM & FUTURE ROADMAP

- Streamlit Real-Time Inference Dashboard Architecture
- Summary of Key Academic & Engineering Contributions
- Research Limitations & Graph Neural Network (GNN) Roadmap

Interactive Demonstration: Streamlit Real-Time Dashboard

Real-Time Inference Architecture

- Lightweight Streamlit UI with dynamic risk probability gauges and visual status cards.
- Instant feature extraction engine transforming raw user inputs into the 14-dimension engineered vector.
- Pre-loaded serialized XGBoost and Random Forest pipelines via Joblib.
- Sub-millisecond single-transaction inference latency (< 1ms).

Demonstrated Simulation Scenarios

- Scenario 1 (Normal Transfer): Standard transaction with valid balances -> Model Output: SAFE (Risk Probability < 0.1%).
- Scenario 2 (Account Drain): High-amount transfer leaving zero origin balance -> Model Output: HIGH RISK FRAUD (Risk Probability > 99.8%).
- Batch CSV Scoring: Allows bulk processing and compliance report exports.

Summary of Contributions & Future Research

Key Project Contributions

- Leak-Free Protocol: 80/20 stratified partition preventing synthetic data contamination.
- Domain Engineering: 14 synthesized features capturing > 97% of tree split gains.
- Superior Benchmark: F1-Score = 99.73% and Recall = 99.51% (missing only 8 out of 1,643 frauds).
- Operational Deployment: Ready-to-use Streamlit dashboard for real-time risk screening.

Future Research Roadmap

- Graph Neural Networks (GNN): Deploy GCN / GraphSAGE to uncover structured money-laundering subgraphs.
- Streaming Ingestion: Integrate Apache Kafka / Flink for distributed sub-second streaming transactions.
- Explainable AI (XAI): Integrate SHAP / TreeExplainer for regulatory compliant transaction audit logs.

THANK YOU FOR YOUR ATTENTION!

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KEY ACADEMIC REFERENCES & PROJECT REPOSITORY

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- [2] T. Chen, C. Guestrin. "XGBoost: A Scalable Tree Boosting System", Proc. 22nd ACM SIGKDD, 2016.
- [3] N. V. Chawla, K. W. Bowyer, L. O. Hall, W. P. Kegelmeyer. "SMOTE: Synthetic Minority Over-sampling Technique", JAIR, 2002.
- [4] L. Breiman. "Random Forests", Machine Learning, 45(1), pp. 5-32, 2001.
- [5] **Project Submodule**: <https://github.com/uit-25730067-chithanh/CS106-Team09-Fraud-Detection>

We cordially welcome all questions, feedback, and discussions from the Committee.